

THE THEORY BEHIND ELECTRICITY

In This Chapter

- Understanding atomic structure
- Harnessing the laws of attraction and repulsion
- Controlling the flow of electrons
- Identifying an element's conductivity and resistance

Electronics is the study of devices that can control the flow of electricity. You can build devices that detect, measure, power, control, count, store, and transmit electricity—and much more. But in order to do all of these things, you first need to know what electricity is and how it flows.

To get to the essence of electricity, you must delve into some of the most basic concepts in physics: atoms and their structure.

Atoms and Their Structure

An atom consists of a cloud of negatively charged electrons surrounding a dense nucleus that contains positively charged protons and electrically neutral neutrons. The relationship between an atom's charged particles—its protons and its electrons—is the key to electricity (much more on this in the following sections of this chapter). Atoms are basic units of matter.

Matter refers to any physical substance; in other words, matter is anything that has mass (measurable stuff) and volume (measurable occupation of space).

Ever wonder about the difference between an electric-powered device and an electronic device? It comes down to a matter of language and general usage. Most of us think of things with basic controls only—such as a lamp, iron, or fan—as electric appliances. Devices with more complex control are viewed as electronic devices.

In this book, we first consider most basic controls such as switches and fuses, and then look into how other electronic components use electricity to perform more and more sophisticated functions.

A chemical element is pure matter consisting of only one type of atom. Every element is composed of an atom with a particular atomic structure that defines it; for instance, the element carbon is composed exclusively of carbon atoms. Elements are ranked by their atomic number on the *periodic table of chemical elements*. The atomic number indicates the number of protons in each atom.

The standard model of an atom has an equal number of protons, neutrons, and electrons but this isn't always the case. The number of neutrons can vary, and each variation is a different isotope of that element. We call the combined number of protons and neutrons nucleons. For example, Carbon-14 is an isotope of Carbon. It has six protons and eight neutrons. It is still Carbon, but the variation in the number of neutrons affects some of its properties.

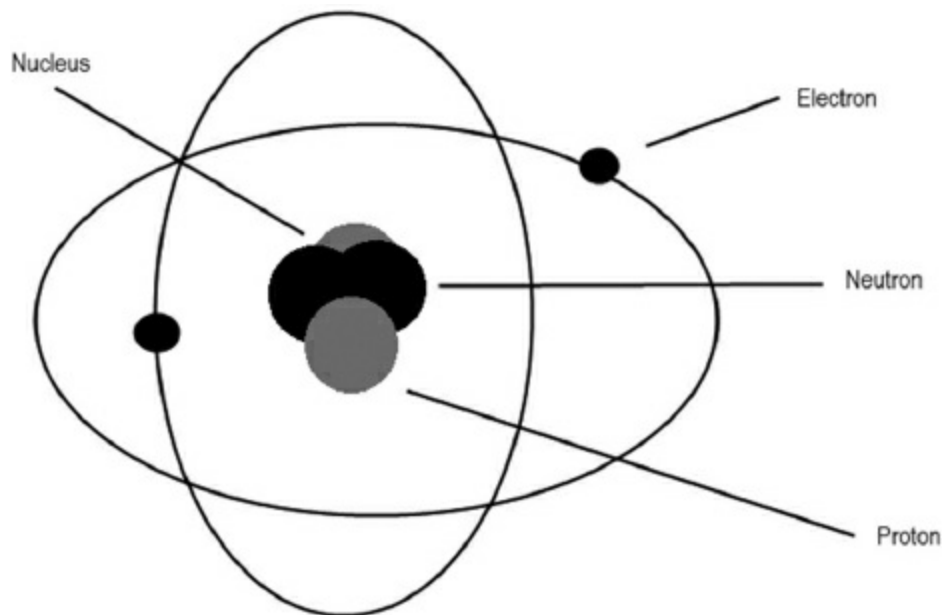


FIGURE 1.1 A carbon atom.

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
		57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
		89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

FIGURE 1.2 A carbon atom.

The **periodic table of chemical elements**, often simply called the periodic table, lists the 118 known elements and basic information—atomic number, relative atomic mass (also known as atomic weight), symbol, and other information, depending on the table—about each element.

Electrons

The atomic number of an element indicates the number of protons. For an electrically neutral or stable atom the number of protons and electrons are equal, which means that once you know the atomic number of an element you know the number of electrons it has. Electrons travel around the nucleus of the atom in an area known as a *shell*. Shells are layered outward from the nucleus. Each shell can hold up to a maximum number of electrons. The innermost shell can hold 2 electrons, the second shell can hold 8, the third shell can hold 18, and the fourth can hold 32.

The following table shows the electron arrangements for some common elements:

Number of Electrons in Various Elements		
Atomic Number	Element Name	Electrons per Shell
1	Hydrogen (H)	1
2	Helium (He)	2
6	Carbon (C)	2,4
7	Nitrogen (N)	2,5
8	Oxygen (O)	2,6
9	Fluorine (F)	2,7
10	Neon (Ne)	2,8
11	Sodium (Na)	2,8,1
12	Magnesium (Mg)	2,8,2
13	Aluminum (Al)	2,8,3
14	Silicon (Si)	2,8,4
15	Phosphorus (P)	2,8,5
16	Sulfur (S)	2,8,6
17	Chlorine (Cl)	2,8,7
19	Potassium (K)	2,8,8,1
20	Calcium (Ca)	2,8,8,2
24	Chromium (Cr)	2,8,13,1
26	Iron (Fe)	2,8,14,2
28	Nickel (Ni)	2,8,16,2
29	Copper (Cu)	2,8,18,1
30	Zinc (Zn)	2,8,18,2
33	Arsenic (As)	2,8,18,5
36	Krypton (Kr)	2,8,18,8
47	Silver (Ag)	2,8,18,18,1
50	Tin (Sn)	2,8,18,18,4
53	Iodine (I)	2,8,18,18,7
79	Gold (Au)	2,8,18,32,18,1
80	Mercury (Hg)	2,8,18,32,18,2
82	Lead (Pb)	2,8,18,32,18,4
92	Uranium (U)	2,8,18,32,18,8,2

Valence Shell

The outermost shell of an atom is known as the *valence shell* (or *valence band*), and

the electrons that inhabit that outer shell are called valence electrons. The more full the valence shell, the less likely it is that an atom will lose electrons when a force is applied. The less full the valence shell, the more likely it is to lose electrons when a force is applied.

Let's compare two elements. As you can see from the preceding table, neon has a full valence shell, meaning that it is unlikely to gain or lose electrons. Copper, on the other hand, has just 1 electron in its valence band, which can hold 32 electrons. This lone electron filling the valence shell is easily attracted away to a nearby atom that has room on its valence shell.

If a valence shell loses or gains an electron, the atom becomes an *ion*. An ion is an atom with a charge. An atom that has more protons than electrons has a positive charge. An atom with more electrons than protons has a negative charge. Because of *electromagnetic force*, negatively charged electrons will leave their own valence shell to travel to another atom that has a positive charge.

Electromagnetic force (emf) is that attraction between positive and negative charges and the repulsion of like charges. It is the basis of interaction between the protons and electrons within atoms holding them together, and the attraction between atoms that have negative and positive charges.

Here's where electricity enters the picture: The movement of electrons on the valence shell when leaving or joining another atom creates electrical current, or electricity. The movement of electrons (and therefore electricity) relies on the two basic concepts that result from electromagnetic force: 1) opposite charges are attracted to each other; and 2) like charges repel each other.

In his pioneering work on electricity, Benjamin Franklin described something that produced electricity as *positive* (positive because it gave current) and the recipient material of that current as *negative* (because it was receiving the electrical charge). This is called conventional theory—the early belief that current traveled from positive to negative.

Today we know now that the opposite is true: Current travels from negative to positive. This is called *electron theory*.

Confusingly enough, many diagrams that are used to describe circuits show the flow of current in *conventional notation*, with current flowing from the positive terminal of a battery to the negative terminal. Others use the more accurate *electron notation*. History creates traditions; we sometimes have to learn to go with the flow!

Conductors, Insulators, and Semiconductors

Some atoms are more stable (or neutral) than others. Stable atoms have an equal number of positively charged protons and negatively charged electrons. The attraction between protons with positive charges and electrons with negative charges holds the atom together unless a force is introduced to separate them.

Conductivity is the tendency of a material to allow the free flow of electrons. *Resistance* is the opposite; it is the tendency of a material to resist the flow of electrons. When we measure conductivity, we refer to it as resistance. A good conductive material is simply said to have very low resistance. The conductivity of a material is determined by how full or empty the valence shell of its atoms is.

An atom with a full valence shell is not going to accept extra electrons, while an atom with a nearly empty valence shell will be able to shed and receive electrons. This flow of electrons among atoms is electricity. As we mentioned previously, copper's nearly empty valence shell allows it to shed and accept electrons, so it is a good conductor of electricity. Neon, with its full valence band, is very nonreactive, so it is resistant to the flow of electricity; in other words, it is an insulator.

Knowing both of these qualities is important to understanding electronics. Electronics relies on our having the ability to control the flow of electricity. We need to be able to slow it, block it, and even modulate it. (More on that later!) This requires that we understand which materials are conductors (highly conductive or low resistance), which are insulators (poor conductors or strong resistance), and which are semiconductors (in between low and strong resistance).

Conductors

Elements that are grouped on the left side of the periodic table have fewer electrons in their valence shell and can serve as good conductors. That's because these electrons are loosely bound to their nuclei (the plural of *nucleus*) and can easily be separated from their atom and travel to a positively charged ion. In other words, these elements allow for electricity—which is simply the flow of electrons—to flow easily. Examples of common metals that are relatively good conductive materials are silver (Ag), gold (Au), and copper (Cu), all of which contain just one electron in their valence shell; that lone electron is easily removed when electricity is flowing. Moving to the right from these metals to the far right of the periodic table you encounter more stable elements that are less conductive.

Insulators

Elements that have full or nearly full valence shells either hold on to their existing electrons or attract electrons so that their valence shell becomes full. These elements are insulators that have great resistance and can slow or block the flow of electricity. They don't have room on their valence shell to accept electrons, and their nearly full outer shell holds tightly to the electrons it already has.

The elements at the far right of the periodic table are called the noble gases. These are extremely good insulators as they are very nonreactive. The naturally occurring noble gases are Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), and Radon (Rn).

Semiconductors

The elements in between the metals and the noble gases on the periodic table are generally semiconductors. Some elements commonly used as semiconductors are silicon (Si) and germanium (Ge). These elements can be combined with others to introduce impurities that can conduct electricity. This process is called *doping*, and when an element is used in this capacity, it is referred to as a *dopant*.

Consider an atom of silicon, which has four electrons in its valence shell. When you look at multiple atoms of silicon, as shown in the following figure, you can see that they arrange themselves quite neatly into what is called a *crystalline structure*, meaning the atoms form a repeating pattern in each direction, with each of its electrons in the valence shell perfectly paired with its neighboring atom.

Now let's look at what happens when two very different dopants are added. Phosphorus (P) has 15 electrons—2 on its inner shell, 8 on its second shell, and 5 on its valence shell. When it bonds with silicon, the combination yields a loosely attached electron. Because that electron can be easily released, a negative charge can easily flow through the doped semiconductor. Phosphorus acts as a donor impurity, because when it is added to silicon it releases or donates electrons. This yields what is called an *n-type* semiconductor, where *n* means negative.

You can create a p-type semiconductor—p meaning positive—by adding boron (B) to silicon. Boron has five electrons, two on its inner shell and three on the valence shell. When you combine these two elements, the bond between the two elements produces a valence shell with seven electrons. This nearly full valent shell does not want to release electrons. However, it does have room to accept an electron in the remaining space, which is referred to as a hole.

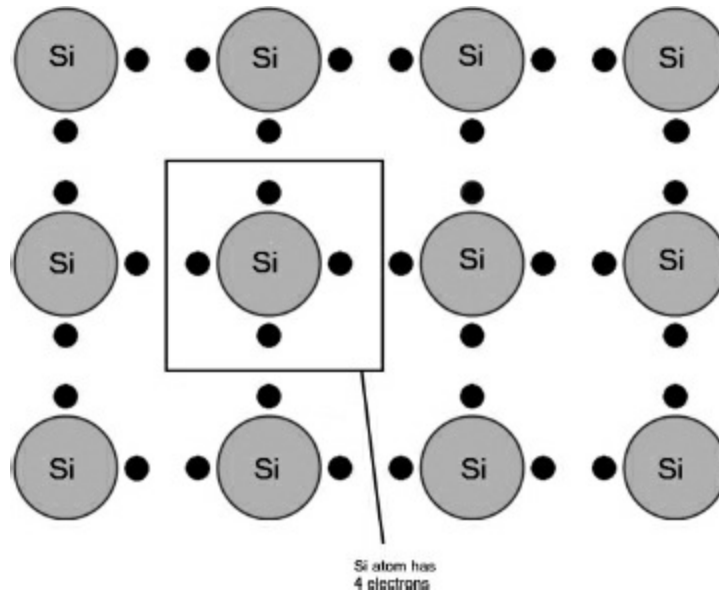


FIGURE 1.3 Atoms of silicon in a crystalline structure. Note that the image doesn't depict all of the atoms on the outside rows; silicon has four electrons in its valence shell.

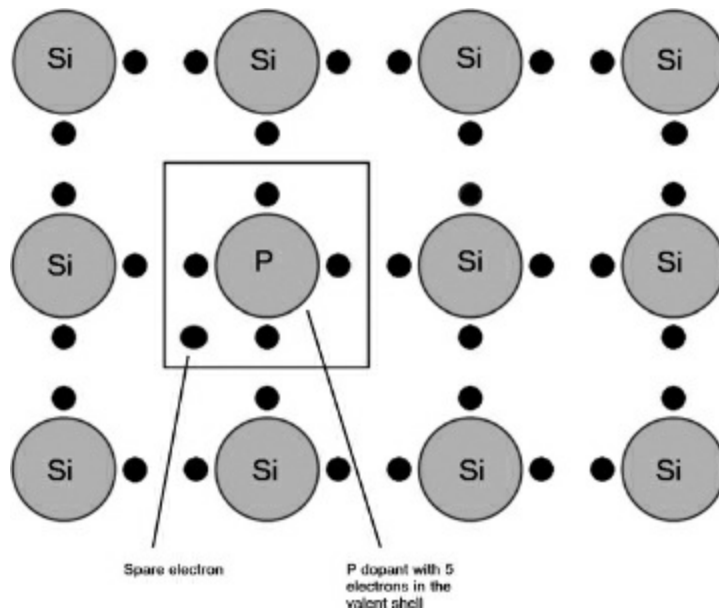


FIGURE 1.4 An n-type semiconductor formed by doping with phosphorus.

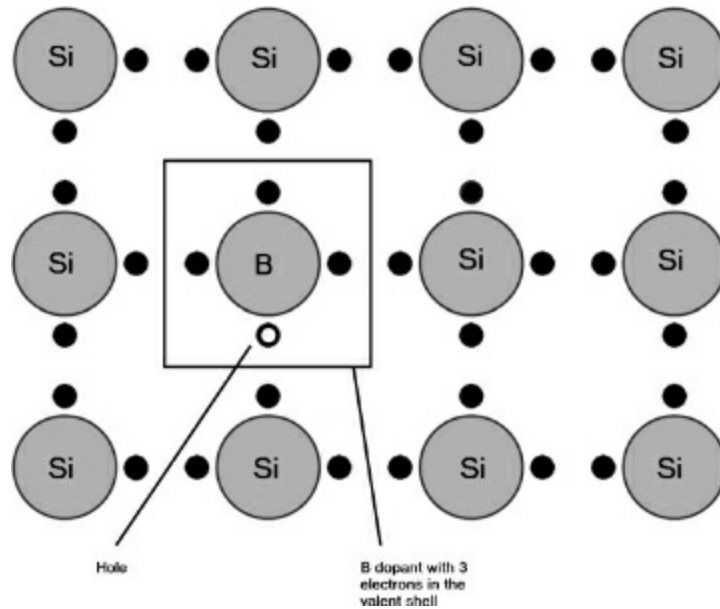


FIGURE 1.5 B dopant with three electrons in the valence shell.

Although the atomic qualities of a material are the most important in determining its conductivity, other factors need to be considered when determining an element's conductivity:

- The physical characteristics of the material: A thick strip of aluminum will conduct more electricity than a thin one. A short wire
- shortens the distance needed for the current to travel compared to a longer wire.
- Temperature: Different materials change in their conductivity depending on the temperature. Metals tend to become less conductive when heated and some become superconductive at extremely low temperatures.

Electron Flow Versus Hole Flow

When an electron leaves an atom, it creates a gap for the next electron to jump into. The electrons move in one direction, so the gaps always open up in the reverse direction.

The movement of the electrons is called electron flow. The opening up of the gaps is called hole flow.

The flow of electrons is like a flow of marbles through a straw. One electron moves

into the space created by the movement of the previous electron down the line.

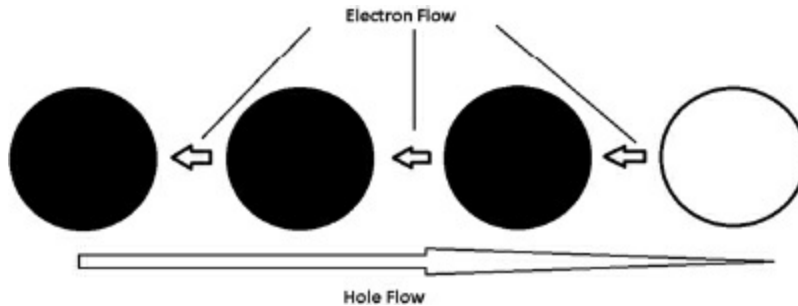


FIGURE 1.6 When electrons move to the empty holes during electron flow, the “movement” of the empty holes is called hole flow.

The Least You Need to Know

- An atom has a nucleus containing positively charged protons and neutrally charged neutrons; the nucleus is surrounded by a cloud of negatively charged electrons.
- A stable atom has equal numbers of protons and electrons. When an atom gains or loses an electron, it becomes an ion—a charged atom.
- Electrons travel in shells or bands around the nucleus. The outer shell is called the valence shell. When electrons move from one atom to another, they create an electric current.
- Materials are classified as conductors, insulators, or semiconductors based on their resistance to conductivity. Semiconductors can be doped to create n-type or p-type semiconductors.

Chapter Review Questions:

1. Electrons are _____-charged particles.
2. The outermost shell (or band) of electrons is called the _____ shell/band.
3. A(n) _____ is an atom with a charge.
4. _____ is the tendency of a material to resist the flow of electrons.
5. Beyond the atomic structure of a material, name two things that affect the conductivity of a material.
6. True/ False: Silver, gold, and copper all have just one electron in their valence shell, these metals are generally considered good insulators.
7. N-type semiconductors are doped to allow for the easy movement of a _____

charge.

8. True/ False: Hole flow is in the same direction as electron flow.
9. The naturally occurring noble gases are: (name at least 3), _____, _____, and _____.
10. _____ charges repel, while _____ charges are attracted to each other.